

DS2500 Project Report - Wildfires in California

Problem Statement and Background

The recent LA fires drew worldwide attention after forcing 200,000 people to evacuate and burning nearly 50,000 acres. Wildfires in California have become increasingly frequent and destructive, stressing the need to understand their primary causes and how these causes have changed over time. This study investigates the causes of wildfires and their relationship to variables such as year, frequency, area-to-perimeter ratio, duration, and fire intensity. This understanding can help agencies such as fire departments and government officials improve mitigation strategies and implement more effective wildfire prevention measures.

Introduction to Data

Collected by the California Department of Forestry and Fire Protection's Fire and Resource Assessment Program (FRAP), this dataset contains 22,261 rows and 21 columns, recording wildfires in California from 1878 to 2023. In a CSV (Comma-Separated Values) format, the file titled “California_Fire_Perimeters_(all).csv” includes data from agencies like the U.S. Forest Service, the Bureau of Land Management, and the National Park Service. The data covers both public and private lands in California, providing details on the following: name, size, cause, alarm date, containment date, and year, as well as the incident number, management agency, fire complex (if applicable), and GIS-based measurements of the fire perimeter, including area and length.

When looking at wildfire data, it's important to recognize what it doesn't show. Numbers can tell us the size of a fire, how long it burned, and what started it, but they don't capture the full impact: homes *destroyed*, families *displaced*, ecosystems *lost*, and the *long road to recovery*. The damage isn't just physical or economic; it disrupts lives in ways that are hard to measure. And if we focus solely on the numbers, we risk overlooking the communities that are hit the hardest and struggle the most to rebuild. Our dataset may have several limitations that could affect the accuracy of our analysis. Missing data, like

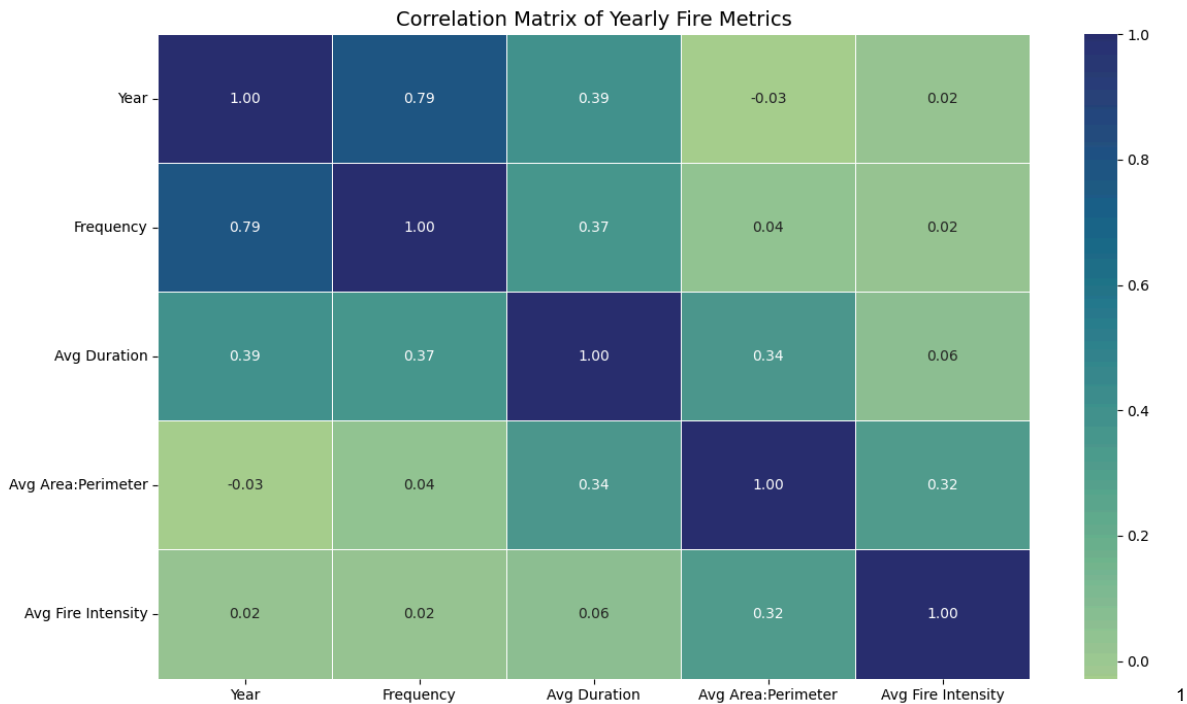
perimeters from lost or damaged historical records, can create an incomplete picture of past fire trends. Smaller fires that don't meet the minimum size criteria may be excluded, recent fires after 2023 aren't reported, and duplicate records could overstate the true frequency of wildfires, all of which can skew the data set's accuracy. Particularly with larger or older fires, some unburned areas may be misclassified as burned, overestimating the size of some wildfires. Lastly, a large number of fires are categorized as having 'Unknown/Unidentified' or 'Miscellaneous' causes, which leads to uncertainty in our analysis, limiting our ability to identify the most common risk factors.

Preparing the Data

We began by cleaning the dataset, removing columns that weren't relevant and dropping rows with missing or blank values in key fields like fire duration, year, area, perimeter, and cause. Records with unclear or unknown causes were also excluded to ensure more accurate analysis. Although the dataset includes wildfire records dating back to 1878, many of the earlier records were removed due to missing or incomplete information. As a result of this cleaning process, our cleaned dataset only included records from 1921 to 2023. Following this, we calculated the duration of each fire using the start and end dates, then removed the original date columns to simplify the dataset.

To better understand the behavior of these wildfires better, we added new features. One measured the area-to-perimeter ratio, giving insight into how spread out or compact each fire was. Another estimated the intensity of each fire by dividing the area burned by the duration, helping us understand how quickly fires were spreading. We also converted the numeric fire cause codes into descriptive text labels to make the data easier to interpret.

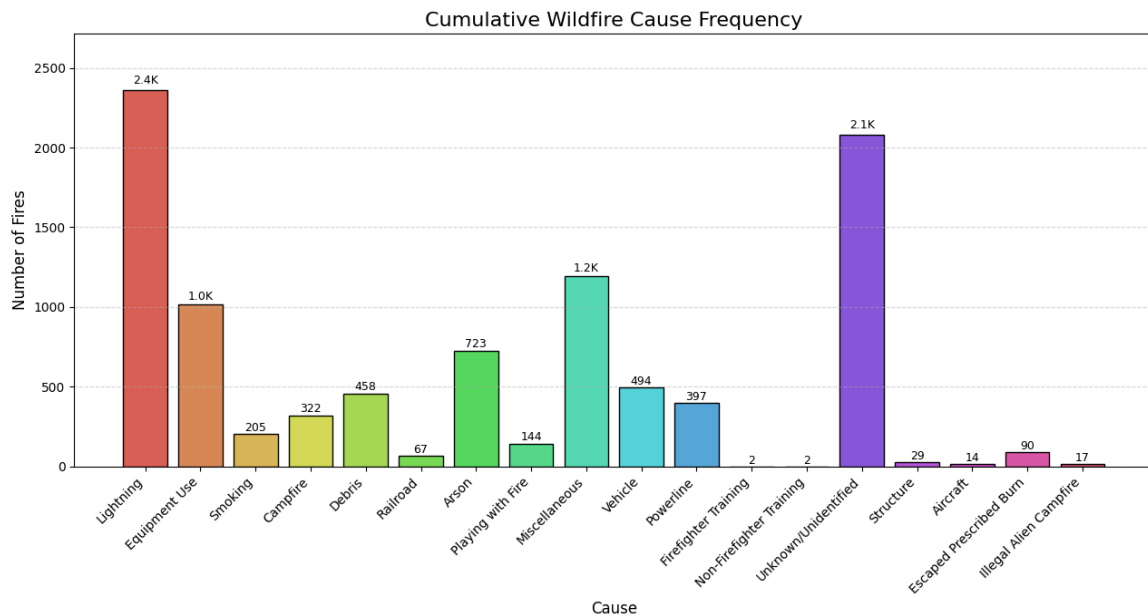
Data Science Approaches



As seen in Fig. 1, the first visualization is a correlation heatmap that explores relationships between five wildfire-related metrics: Year, Frequency, Average Duration, Average Area-to-Perimeter Ratio, and Average Fire Intensity. The strongest correlation is between Year and Frequency, with a positive value of 0.79 indicating that wildfire occurrences have generally increased over time. There is also a moderate positive relationship between Average Duration and Frequency (0.37), suggesting that fires may burn longer as they become more frequent. Additionally, Average Duration has a moderate positive correlation with the Area-to-Perimeter Ratio (0.34), meaning that longer-lasting fires might spread out more and have a messier, less compact shape. In contrast, the correlations between Area-to-Perimeter Ratio and both Year and Frequency are relatively weak. This makes sense, as how spread out a fire is doesn't necessarily depend on when it happened or how often fires occur. However, the correlation between Area-to-Perimeter Ratio and Fire Intensity (0.32) shows that more irregularly

¹ Correlation matrix of wildfire metrics including Year, Frequency, Average Duration, Area-to-Perimeter Ratio, and Fire Intensity.

shaped fires might burn a bit stronger. On the other hand, most of the other factors don't seem to have much to do with fire intensity, which suggests that how powerful a fire is might depend on completely different things than how big or frequent it is.



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As seen in Fig. 2, the second visualization shows the frequency of wildfire causes cumulative from 1921 to 2023. The bar chart highlights a wide range of ignition sources, from natural causes like Lightning to human-related activities such as equipment use, arson, campfires, and more. Lightning stands out as the most frequent cause, responsible for around 2,400 fires, followed closely by unknown/unidentified causes and miscellaneous. Less frequent but still notable causes include vehicle use, powerlines, and escaped prescribed burns. This visualization helps identify which causes are most prevalent.

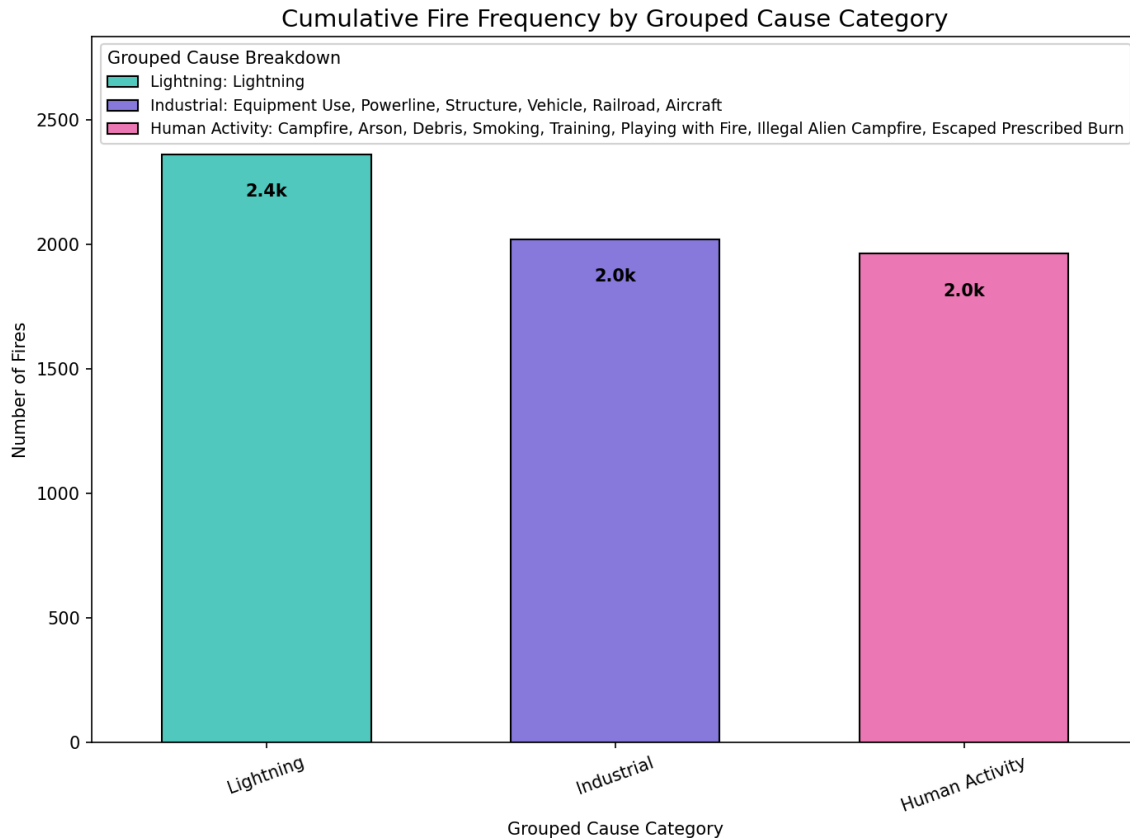
To classify the cause of wildfires, we focused on supervised machine learning algorithms that were designed to predict categorical data through classification. We initially explored several classification models, including the K-Nearest Neighbors (KNN) classifier, Decision Trees, and Random

² Bar Chart depicting the total number of wildfires caused by each source from 1920 to 2023.

Forest. After some experimentation, we chose to proceed with KNN and Random Forest, as they offered the most potential given the nature of our data. We did a brief exploration with multiple linear regression to predict wildfire size; however, the model performed poorly, likely due to weak correlations between size and other numerical features, as shown in our correlation matrix.

KNN is a straightforward algorithm that classifies a data point based on the majority class of its closest “neighbors” in the dataset. We started by testing it with features like fire duration, shape area, and shape perimeter length, but the model’s accuracy was too low to be considered reliable (20-30%), even after adjusting the number of neighbors and experimenting with different distance metrics like Manhattan and Euclidean. We later expanded our feature set to include log-transformed versions of skewed variables and new features like area-to-perimeter ratio and fire intensity. Random Forest, a model that builds multiple decision trees and aggregates their results for more robust predictions, also saw poor initial performance. To improve both models, we tuned hyperparameters like maximum tree depth and the number of estimators for Random Forest, and tested different values of “k” and distance metrics for KNN.

After many iterations and disappointingly low accuracy scores, we realized that the issue might lie more with the data itself than the models. A large portion of our dataset had vague or unhelpful labels like “Unknown” or “Miscellaneous” for wildfire causes. While these categories might be used in real-life reporting when the exact cause isn’t known, they lacked the necessary detail for our analysis. Removing those records led to a noticeable improvement in accuracy, with both models performing about 10% better in terms of accuracy. Still, many cause categories had very few examples, making them difficult for any model to learn. So, we reorganized the target variable by grouping similar causes into broader categories. Our first attempt involved groupings of the causes into broader categories including Transportation, Industrial, Lightning, Accidental Human Activity, and Intentional Human Activity. This step improved model accuracy to around 50%.

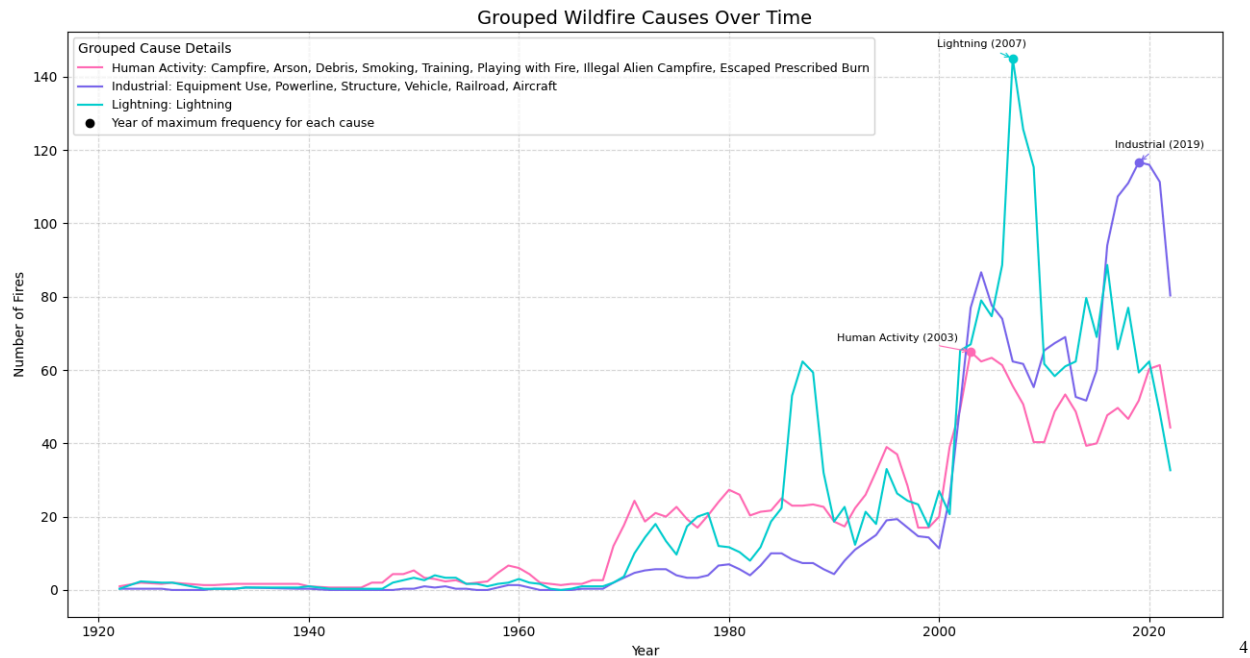


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As seen in Fig. 3, we simplified the groups even further and settled on the following three categories: Lightning, Industrial (e.g., powerlines, vehicles, equipment), and Human Activity (both accidental and intentional). By grouping the causes into three categories, it allowed our KNN model to predict the cause more accurately with the given set of features. It increased our KNN accuracy from 50% to 63% and our Random Forest classifier accuracy from 52% to 64%.

Initially, the model struggled because there were too many distinct causes, many of which had very few examples. This made it difficult for the algorithm to learn meaningful patterns. By narrowing it down to just three broader categories, we reduced complexity and gave the model a better chance at making accurate predictions.

³ Bar chart showing the frequency of wildfires for each grouped cause category.



To better understand how different wildfire causes have changed over time, we created a line graph showing the frequency of fires for each grouped cause (Human Activity, Industrial, and Lightning) from 1921 to 2023. Fig. 4 helped us visualize the trends across different decades and see how certain causes became more or less common. We also highlighted the peak year for each cause directly on the graph to make it easier to spot when each type of fire was at its highest. It gave us a clearer picture of how wildfire patterns have changed over the years and which causes stood out the most during certain periods.

Results and Conclusions

Through this project, we learned how to optimize supervised machine learning to best classify wildfire causes and explored how fire characteristics like duration, frequency, and intensity relate to one another. After testing several models and adjusting both features and target categories, we found that grouping causes into three broader categories (Human Activity, Industrial, and Lightning) yielded the best results, with accuracies of 0.63 for KNN and 0.64 for Random Forest. These results suggest the models could be useful in real-world scenarios, especially for identifying likely causes when labels are missing,

⁴ Line graph showing the frequency of fires by grouped cause over time, with the peak year marked for each.

as was the case for many entries in our dataset. While we did not predict causes for these unknown entries due to the lack of verification labels, the models show promise for future use in similar situations.

Our visualisations gave us a clearer picture of what's happening with wildfire causes. The bar chart indicated that lightning is the most frequent cause, with around 2.4k cases, while industrial and human activity each account for about 2.0k. As we mentioned earlier, the correlation matrix revealed a strong positive relationship (0.79) between year and fire frequency, indicating that wildfires have become more frequent over the years. This points to a larger issue: we need stronger policies and more serious efforts to address climate change, which may be contributing to this increase of wildfires. It also reflects how careless and irresponsible our actions have been. The line graph reinforces this shift. In 2007, lightning was the leading cause, but in recent years, industrial activity has become more dominant. This change proves just how much of the problem is now driven by human-driven development.

Based on these results, the real-life implementation of our model could be beneficial for firefighters, helping them to implement more effective mitigation strategies tailored to the identified cause of a wildfire.

Future Work

For future work, one of our main goals is to find a more detailed dataset, ideally one that includes geographical information such as coordinates. This would allow us to explore spatial patterns in wildfire causes and outcomes. We also hope to revisit multiple linear regression, this time using it to estimate the size of a wildfire based on numerical features like duration, intensity, and possibly environmental factors such as temperature and wind speed that would be included in a more detailed dataset. Another direction would be to use the data to predict the most effective mitigation strategies by analyzing which fire characteristics (e.g., duration, size) respond best to certain interventions. Lastly, implementing a time series analysis could help us predict future wildfire trends based on past patterns. This would make it easier to plan ahead and be better prepared for future fires.

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